

Total No. of Questions : 9]
(2042)

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**BCA (CBCS) RUSA IInd Semester
Examination**

3743

MATHEMATICS-II

Paper : BCA-0201

Time : 3 Hours]

[Maximum Marks : 70

Note :- Attempt *one* question from each Unit. Q. No. 9 is compulsory.

Unit-I

1. (a) Verify Lagrange's mean value theorem for the function $f(x) = (x - 2)(x - 3)(x - 4)$ in the interval $[0, 4]$. Also find the value of 'C'.

- (b) If $y = (ax + b)^m$ and $y_n = \frac{d^n y}{dx^n}$, show that :

$$y_n = m(m-1)(m-2) \dots (m-n+1)$$

$$(ax + b)^{m-n} a^n$$

2. (a) By using mean value theorem find the approximate value of $\sqrt{66}$.
- (b) Apply Leibnitz theorem to find the third order derivative of :

$$f(x) = (x^2 + 1)e^{2x} \text{ at } x = 0 \quad 5,5$$

Unit-II

3. (a) Find the greatest common divisor of 275 and 200 and express it in the form $m.275 + n.200$.
- (b) Show that the relation of divisibility in the set of integer is reflexive, transitive but not symmetric. 5,5
4. (a) Prove that if a and b be two integers, then $a \equiv b \pmod{m}$ if and only if a and b has the same remainder when divided by m .
- (b) For positive integer a and b show that :
- $$\text{GCD}(a, b) \times \text{LCM}(a, b) = (a \times b) = ab \quad 5,5$$

Unit-III

5. (a) Show that the set of all 2×2 non-singular matrices form an infinite non-abelian group under the composition of matrix multiplication.

(b) Show that the set $G = \{1, \omega, \omega^2\}$ of cube roots of unity form a finite abelian group of order 3 under multiplication of complex number. 5,5

6. (a) Let $M_g(I) = \left\{ \begin{bmatrix} a, b \\ c, d \end{bmatrix} \right\}$ where $a, b, c, d \in I$, show that $M_g(I)$ form a monoid under the operation of matrix multiplication.

(b) Define a cyclic group and prove that every cyclic group is abelian but converse is not true. 5,5

Unit-IV

7. (a) Prove that the set of integer is a ring with respect to usual addition and multiplication.

(b) Prove that $\langle R, +, . \rangle$ where R is set of all reals, is a commutative ring with unity. 5,5

8. (a) Let R be a ring. Then show that the following conditions are equivalent :

(i) R has no zero divisor

(ii) R satisfies left cancellation law

(iii) R satisfies right cancellation law

- (b) Show that the set of rationals 'Q' is a field under composition of addition $\oplus$ and multiplication $\odot$, given as :

$$a \oplus b = a + b - 1 \quad \text{and} \quad a \odot b = a + b - ab \quad 5,5$$

Unit-V

(Compulsory Question)

9. (i) Discuss applicability of Rolle's theorem for the function $f(x) = |x|$ in $[-3, 3]$.
- (ii) If $f(x) = x^2 \sin 2x$, find the value of $f'''(0)$.
- (iii) If a/b and a/c , then show that $a/(bx + cy) \forall x, y \in \mathbb{Z}$.
- (iv) Solve the following congruence $3x \equiv 4 \pmod{5}$.
- (v) Find the LCM of (272, 1479).
- (vi) Define a group.
- (vii) Show that the set $G = \{1, 2, 3, \dots, n-1\}$ does not form a group under multiplication modulo n , where ' n ' is a composite number.
- (viii) Let R bearing such that $x^2 = x \forall x \in R$, show that R is a commutative ring.
- (ix) Define a ring and a ring with unity.
- (x) Prove that every field is an integral domain.